

SMe Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y} = (Y_1, \dots, Y_n)$ Sample Data: $\mathbf{y} = (y_1, \dots, y_n)$ $f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_i f_{Y_i}(y_i; \theta)$, if y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \leq y_1, \dots, Y_n \leq y_n)$
Parameter Space: Θ CDF: $F_{\mathbf{Y}}(\mathbf{y})$ PDF/PMF: $f_{\mathbf{Y}}(\mathbf{y})$ or $p_{\mathbf{Y}}(\mathbf{y})$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i) \mathbb{P}(Y = y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y) f_Y(y) dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \subseteq \Theta\}$ is model.

Likelihood Function: $L(\theta; \mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$ **MLE:** $\hat{\theta}(\mathbf{y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{y})] = [U(\hat{\theta}) = 0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta; x) \propto L(\theta; y)$ [Strong: $L_f(\theta; x) \propto L_g(\theta; z)$; f, g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y; \theta_1) = f(y; \theta_2), \forall y$ then $\theta_1 = \theta_2$ (i.e. $\text{map} : \theta \rightarrow f(y; \theta)$ is 1-1)

• Method: 1.By def $\Rightarrow$ ratio not depend on $y \Rightarrow \theta_1, \theta_2$ relationship. (with any additional constraint) 2.MLE not unique $\Rightarrow$ not identifiable 3.Give an example θ 4.Fisher information

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix $\mathbf{M}$: $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T \mathbf{M} \mathbf{z} > 0 [\geq 0] \Leftrightarrow (> 0)^1 n \text{ positive eigenvalues. } ^2 \text{ full rank. } ^3 \text{ invertible}$

Score Func / Vector $\mathbf{U}$	Single: $U(\theta, \mathbf{Y}) = \ell'(\theta; \mathbf{Y})$	$\mathbb{E}[U] = \mathbf{0}$	$Var[U] = \mathbf{I}(\theta)$
	Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1} \ell(\theta; \mathbf{Y}), \dots, \partial_{\theta_p} \ell(\theta; \mathbf{Y})]^T \in \mathbb{R}^p$	$\mathbb{E}[\mathbf{U}] = \mathbf{0}$	$Var[\mathbf{U}] = \mathbf{I}(\theta) \quad I_{j,k} = -Cov(\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j}, \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}) = -\mathbb{E}[\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j} \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}]$
CLT of $\mathbf{U}$	Single: $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, \mathbf{I}(\theta_0)), n \rightarrow \infty$		
	Multiple: $\mathbf{U}(\theta_0; \mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, \mathbf{I}(\theta_0)), n \rightarrow \infty$	Also: $\mathbf{U}(\theta_0; \mathbf{Y})^T \mathbf{I}^{-1}(\theta_0) \mathbf{U}(\theta_0; \mathbf{Y}) \sim \chi_p^2$	
Fisher Information $\mathbf{I}$	Single: Observed: $J(\theta, \mathbf{y}) = -U'(\theta, \mathbf{y}) = -\ell''(\theta; \mathbf{y})$	Expected: $\mathbf{I}(\theta) = -\mathbb{E}[\ell''(\theta; \mathbf{Y})]$ Unit: $i(\theta) = \mathbf{I}(\theta)/n = -\mathbb{E}[\ell''(\theta; Y_i)]$	
	Multiple: Observed: $\mathbf{J}(\theta; \mathbf{y}) \in \mathbb{R}^{p \times p}, J_{j,k} = -\frac{\partial^2 \ell(\theta, \mathbf{y})}{\partial \theta_j \partial \theta_k}$	Expected: $\mathbf{I}(\theta) \in \mathbb{R}^{p \times p}, I_{j,k} = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ ps: $\mathbf{I}(\theta)$ always positive semi-definite	
$\mathbf{I} \uparrow \Rightarrow \mathbf{Y}$ 对 θ 越敏感; 信息量大; 参数估计更精确		A model is <i>identifiable</i> iff $\mathbf{I}(\theta)$ is positive definite. ps:(一维) identifiable if $\mathbf{I}(\theta) \neq \mathbf{0}$	
CLT of MLE CLT of $\mathbf{l}$	Single: $\mathbf{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, \mathbf{I}^{-1}(\theta_0)), n \rightarrow \infty$	$\widehat{Var}(\hat{\theta}) \approx \mathbf{I}^{-1}(\hat{\theta})$	$ESE(\hat{\theta}) \approx \sqrt{\mathbf{I}^{-1}(\hat{\theta})}$ estimated standard error
	Multiple: $\mathbf{I}(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}_p(\theta_0, \mathbf{I}^{-1}(\theta_0)), n \rightarrow \infty$	Also: $(\hat{\theta} - \theta_0)^T \mathbf{I}(\theta_0) (\hat{\theta} - \theta_0) \sim \chi_p^2$	

2.1 For Single Parameter

*Random Variable / Function: $L(\theta; \mathbf{Y})$ is random function (NOT pdf). For fixed θ , it's r.v. $\hat{\theta}(\mathbf{Y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{Y})]$ is r.v. ps: $\mathbf{I}^{-1}, \mathbf{J}^{-1}$ 是指其负一次方, 不是反函数 True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, \mathbf{I}(\theta_0))$ $T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{\mathbf{I}(\theta_0)}} \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 不用 MLE, 泰勒展开	$\left\{ \theta : \frac{U^2(\theta)}{\mathbf{I}(\theta)} < z_{\alpha,1}^2 \right\}$	or $T_{Score}^2(\mathbf{Y}) \sim \chi_1^2$ $T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{\mathbf{I}(\hat{\theta})}}, \frac{U(\theta_0)}{\sqrt{\mathbf{J}(\hat{\theta}_0)}}, \frac{U(\theta_0)}{\sqrt{\mathbf{J}(\hat{\theta})}}$
Wald Test or Maximum Likelihood(ML) Test	$T_{Wald}(\mathbf{Y}) = \sqrt{\mathbf{I}(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta} - \theta_0}{\sqrt{\mathbf{I}^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 构建 CI	$\{ \theta : (\hat{\theta} - \theta)^2 \mathbf{I}(\theta) < z_{\alpha,1}^2 \}$	or $T_{Wald}^2 \sim \chi_1^2$; $n \rightarrow \infty, T_{Wald} \simeq T_{Score}$ $T_{Wald}(\mathbf{Y}) \approx \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{I}^{-1}(\hat{\theta})}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{J}^{-1}(\theta_0)}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{\mathbf{J}^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi_1^2$ as $n \rightarrow \infty$ $LR = L(\theta_0)/L(\hat{\theta}), T_{LR} = -2\ln(LR)$	$H_0 : \theta = \theta_0$ 最小 CI, Most Powerful	$\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha,1}^2 \}$	ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \simeq (\theta_0 - \hat{\theta})^2 \mathbf{J}(\hat{\theta})$
$T_{Score} \simeq T_{Wald} \simeq T_{LR}$ as $n \rightarrow \infty$; $J(\theta_0) \approx \mathbf{I}(\theta_0)$ as $n \rightarrow \infty$; When $\ell(\theta)$ 是关于 θ 的二次函数 $\Rightarrow$ All same. Geometric: (T_S , 比斜率), (T_W , 比宽度), (T_{LR} , 比高度) ps: $z_{\alpha,n} = \chi_{\alpha,n}^2$				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{\mathbf{I}(\theta_r)}$
Advantages	Converge <i>very quickly</i> near optimum	$\mathbf{I}$ is constant / easier to calculate; Converge for wider θ_0
Stopping Rule	Given tolerance level ε : ¹ Stop after r if $ U(\theta_r) \leq \varepsilon$ (Common) or ² Stop after $ \theta_r - \theta_{r-1} \leq \varepsilon$	

2.3 For Multiple Parameters

*此节中, θ 代表向量, $\theta = (\theta_1, \dots, \theta_p)$ $\mathbf{I}^{-1}$ 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T_{Score}^2, T_{Wald}^2, T_{LR}$)	Hypothesis	(Confidence Region) CR
Score Test	$\mathbf{U}(\theta_0)^T \mathbf{I}^{-1}(\theta_0) \mathbf{U}(\theta_0) > z_{\alpha,p}$	$\theta = \theta_0$	$\{ \theta : \mathbf{U}(\hat{\theta})^T \mathbf{I}^{-1}(\theta) \mathbf{U}(\hat{\theta}) < z_{\alpha,p} \}$
Wald Test	$(\hat{\theta} - \theta_0)^T \mathbf{I}(\theta_0) (\hat{\theta} - \theta_0) > z_{\alpha,p}$	$\theta = \theta_0$	$\{ \theta : (\hat{\theta} - \theta)^T \mathbf{I}(\theta) (\hat{\theta} - \theta) < z_{\alpha,p} \}$
log-Likelihood Ratio Test	$2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha,p}$	$\theta = \theta_0$	$\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha,p} \}$
$\mathbf{I}(\theta_0) \approx \mathbf{I}(\hat{\theta}), \mathbf{J}(\theta_0), \mathbf{J}(\hat{\theta})$ ps: $z_{\alpha,p} = \chi_{\alpha,p}^2$ T_{LR} : simple hypothesis testing. ($\dim(\mathcal{A}) = 0$) θ 都是向量			

Marginal Normal distribution $\hat{\theta}$:

$\hat{\theta}_j \sim \mathcal{N}(\theta_j, [\mathbf{I}^{-1}(\underline{\theta}_0)]_{jj})$ θ_j : true para

$\cdot ESE(\hat{\theta}_j) = \sqrt{[\mathbf{I}^{-1}(\hat{\theta})]_{jj}}$

$\cdot CI(\theta_j): \hat{\theta}_j \pm z_{\alpha} ESE(\hat{\theta}_j)$

Composite Hypothesis Test: $H_0 : \underline{\theta} \in \mathcal{A}, H_1 : \underline{\theta} \notin \mathcal{A}$ reduced model $LR = \frac{\max_{\theta \in \mathcal{A}} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\hat{\theta}_{\mathcal{A}})}{L(\hat{\theta}_{\Theta})}$ ps: 多维未知参数, H_0 只想其中几个时用 || $\dim(\mathcal{A})$ 常数时为 0

Generalised Likelihood Ratio Test: $T_{GLR} = -2\ln(LR) = 2(\ell(\hat{\theta}_{\mathcal{A}}) - \ell(\hat{\theta}_{\Theta})) \sim \chi_{p-r}^2$ p : # of parameters. $r = \dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0)	*此节中, θ 代表向量	Newton-Raphson Method	Fisher's Method of Scoring
Formula		$\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r) \mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r) \mathbf{U}(\theta_r)$	$\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1} \mathbf{U}(\theta_r) = \theta_r + \mathbf{I}^{-1}(\theta_r) \mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$ If $\mathbf{I} = \text{diag}(a, b, \dots) \Rightarrow \mathbf{I}^{-1} = \text{diag}(a^{-1}, b^{-1}, \dots)$			

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider θ as a r.v.. Bayesian approach $\Rightarrow$ Probability are used to represent degree of believe about the unknown θ
Prior Distribution $p(\theta)$: Prior distribution $p(\theta)$ contains all available prior information about θ *before observing any data*.
· **Hyper-Parameters:** 由于 $p(\theta)$ 是已知,服从某种分布 (with parameter γ). We denoted $p(\theta)$ as $p(\theta; \gamma)$ where γ is *Hyper-parameters*. 定义: $\gamma_{prior} \rightarrow \gamma_{post}$
Likelihood function $p(y|\theta) = f(y|\theta)$: Any observed data is considered as a realisation of r.v. Y with $f(y|\theta)$
Posterior Distribution $p(\theta|y)$: Posterior distribution $p(\theta|y)$, contains all available knowledge.
· **Bayes' Rule / Theorem:** $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{f(y|\theta)p(\theta)}{m(y)}$ $p(y) = m(y) = \int_{\theta \in \Theta} f(y|\theta)p(\theta)d\theta$ (Normalizing constant)

Name	MAP _{Maximum A Posteriori}	MMSE _{Posterior Mean / Bayesian Minimum Mean Squared Error Estimator}	Posterior Median $\hat{\theta}_{median}$
Single Para	$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta y)$	$\hat{\theta}_{MMSE} = \mathbb{E}[\theta y] = \int_{\theta \in \Theta} \theta p(\theta y)d\theta$	$\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta y)d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta y)d\theta$
Name	Posterior Var	$C_{\alpha} = [a, b]$ (1 - α)100% Bayesian Credible Intervals (or Simply Credible Intervals)	
Single Para	$Var[\theta y] = \int_{\theta \in \Theta} (\theta - \mathbb{E}[\theta y])^2 p(\theta y)d\theta$	$\mathbb{P}(a \leq \theta \leq b y) = \int_a^b p(\theta y)d\theta = 1 - \alpha$ ps: $\mathbb{P}(\theta < a y) = \mathbb{P}(\theta > b y) = \alpha/2$	

Bayesian Decision Theory: 目的: ¹Estimating parameter θ ²Test Hypothesis about θ ³Find CI. ps: 后两种有很多种方法来判断
1. Choose a *Loss Function* $Q(\hat{\theta}, \theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)
2. Minimises Posterior Expacted Loss Function: $\hat{\theta}_Q(y) = \arg \min_{\hat{\theta}} \mathbb{E}[Q(\hat{\theta}, \theta)|y] = \arg \min_{\hat{\theta}} \int_{\theta \in \Theta} Q(\hat{\theta}, \theta)p(\theta|y)d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For $\theta, p(\theta)$ is *Conjugate Prior* for $p(y|\theta) = f(y|\theta)$ iff 对 $p(\theta)$ 用 Bayes' Rule 得到的 $p(y|\theta)$ 都在同一类型的分布
Conjugate Family: The collection of all *Conjugate Prior*.
Choice of Prior: Observed data number n : ¹ n small $\Rightarrow$ Posterior shows a great dependency of the choice of prior.
² n middle $\Rightarrow$ posterior less affected by choice of prior. ³ n large $\Rightarrow$ choice of prior has limited effect on posterior.(Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: ¹ $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$ for most cases, especially n large. ² n large: $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}, CI \approx C_{\alpha}$

3.3 Approximate Bayesian Inference

Intractable Posterior: Non-Conjugate model. (not analytic, no closed-form)
Monte Carlo Methods: Any func $g(\theta) || I_g = \int_{\theta \in \Theta} g(\theta)p(\theta|y)d\theta \approx \frac{1}{M} \sum_{i=1}^M g(\theta^{(i)}) \xrightarrow{a.s.} I_g$ $\theta^{(i)} \sim p(\theta|y), i = 1, ..., M$
1. **Importance Sampling (IS):** ¹ Simulating samples from another distribution $q(\theta)$ (*Proposal Distribution*) Consistent, Biased for finite
 $^2 w^{(i)} = \frac{\pi(\theta^{(i)}; y)}{q(\theta^{(i)})}, \pi(\theta; y) = f(y|\theta)p(\theta)$ ³ IS estimator: $\hat{I}_g^M = \sum_{i=1}^M g(\theta^{(i)}) \overline{w}^{(i)} \xrightarrow{a.s.} I_g$, Normalized Weight: $\overline{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^M w^{(j)}}$ $\theta^{(i)} \sim q(\theta)$ ps: $\sum_{j=1}^M \overline{w}^{(j)} = 1$
Remark: Choice of $q(\theta)$ is important. $|| \hat{I}_g^M$ depends on: ¹# of M ; ²mismatch between proposal and the integrand of the integral. Good if $q(\theta) \propto g(\theta)p(\theta|y)$, difficult to achieve.:(
2. **Markov Chain Monte Carlo (MCMC):** MCMC 算法通过 Markov Chain 生成样本, 使得样本接近服从后验分布.(计算量大)
Laplace Approximation: (最简单的估算 posterior 的方法) $p(\theta|y) \approx \mathcal{N}(\underline{\theta}_{MAP}, H^{-1}(\underline{\theta}_{MAP}))$ 类似 MLE 的 J $H(\theta) = -\frac{d^2 \ln(p(\theta|y))}{d\theta^2}$ (适合 $p(\theta|y)$ 接近正态分布的时用)
Variational Inference (VI): ¹Choose $q(\theta; \eta), \eta$ 是 q 的参数. ²选择衡量 $p(\theta|y), q(\theta; \eta)$ 差异的方法 e.g.**KL divergence:** $KL(q(\theta; \eta)||p(\theta|y)) = \int_{\theta \in \Theta} q(\theta; \eta) \ln(\frac{q(\theta; \eta)}{p(\theta|y)})d\theta$
³ 优化参数 (Find parameters to minimise the divergence): $\eta^* = \arg \min_{\eta \in A_{\eta}} KL(q(\theta; \eta)||p(\theta|y))$ ⁴ $p(\theta|y) \approx q(\theta; \eta^*)$
Remark: Both Laplace Approximation & Variation Inference approximate $p(\theta|y)$ by a parameter family of distributions. || 拉普拉斯是正态分布家族,VI 是更普遍分布家族

4 Linear Model

4.1 Simple Linear Model (With One Predictor)

Model: $Y_i = \mu_i + \varepsilon_i, \mu_i = \beta x_i, i = 1, 2, ..., n$ || Suppose ¹ ε_i independent ² $\mathbb{E}[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$ || Y_i response variable, x predictor variable.

Least Square (LS) Estimation	RSS _{Residual Sum of Squares} : $S(\beta) = \sum (y_i - \mu_i)^2 = \sum (y_i - x_i \beta)^2$	LS Method: $\hat{\beta}_i = \arg \min_{\beta} S(\beta)$ ps: 求偏导 = 0 $\hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$
$\mathbb{E}[\hat{\beta}], Var[\hat{\beta}], Var[\widehat{\beta}]$	$\mathbb{E}[\hat{\beta}] = \beta, unbiased Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \sigma^2$ (σ^2 known)	$Var[\widehat{\beta}] = \sigma_{\widehat{\beta}}^2 = (\sum x_i^2)^{-1} \cdot \widehat{\sigma}^2 \widehat{\sigma}^2 = \frac{1}{n-1} \sum (y - x_i \hat{\beta})^2$
H_0, H_1 Test CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / Var(\widehat{\beta}) \sim t_{n-1}$ (σ^2 unknown)
Other	Fitted Values: $\hat{\mu}_i = x_i \hat{\beta}$ Residuals: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i$	For CI / H_0/H_1 Test: Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

4.2 General Linear Model

Linear Model: $Y_i = \mu_i + \varepsilon_i, i = 1, 2, ..., n, \mu = f(\beta) || f(\beta)$ linear. || $\varepsilon_i \stackrel{i.i.d.}{\sim}$ distribution. $\mathbb{E}[\varepsilon_i] = 0, Var[\varepsilon_i] = \sigma^2$
· **Matrix Expression:** $\underline{y} = \underline{X}\underline{\beta} + \underline{\varepsilon} ||$ Response Vector: $\underline{y}$ | Design / Model Matrix: $\underline{X}$ | Parameters / Coefficients: $\underline{\beta}$ | Additive Noise Vector: $\underline{\varepsilon}$
· **Some Defs:** Parameter Estimates: $\hat{\underline{\beta}}$ | Estimated / Fitted Value Mean Vector: $\hat{\underline{\mu}} = \underline{X}\hat{\underline{\beta}}$ | Estimated / Fitted Residual Vector: $\hat{\underline{\varepsilon}} = \underline{y} - \hat{\underline{\mu}}$
Lamma|E, Var in Matrix: If r.v. $\underline{v}, \underline{A}, \underline{b}$ constant $\Rightarrow$ ¹ $\mathbb{E}[\underline{A}\underline{v} + \underline{b}] = \underline{A}\mathbb{E}[\underline{v}] + \underline{b}$ ² $Var[\underline{A}\underline{v} + \underline{b}] = \underline{A}Var[\underline{v}]\underline{A}^T$
Metric Predictor: 设计矩阵用连续观测值. || **Factor Predictor:** 离散数据用指示函数 (0/1) 填满, 每个"小组"对应一个指示函数.

4.3 Linear Model Theory

Orthonormal Matrix $\underline{Q}$	Def: $\underline{Q} :=$ columns and rows are orthonormal vectors. $\underline{Q}^T = \underline{Q}^{-1}$ $\underline{Q}\underline{Q}^T = \underline{Q}^T \underline{Q} = \underline{I}_n$ $\forall \underline{v}, \underline{v} ^2 = \underline{Q}\underline{v} ^2 = \underline{Q}^T \underline{v} ^2$
QR Decomposition	$\forall \underline{X} \in \mathbb{R}^{n \times p} \Rightarrow \underline{X} = \underline{Q} \begin{pmatrix} \underline{R} \\ 0 \end{pmatrix} = \underline{Q}\underline{R}$ $\underline{R} \in \mathbb{R}^{p \times p}$ is upper triangular matrix; $\underline{Q} \in \mathbb{R}^{n \times n}$ is orthonormal matrix; $\underline{Q}$ 前 p 列等于 $\underline{Q} \in \mathbb{R}^{n \times p}$
Least Square (LS)	$RSS := S(\underline{\beta}) = \sum_{i=1}^n \varepsilon_i^2(\underline{\beta}) = \sum_{i=0}^n (y_i - \mu_i(\underline{\beta}))^2 = \underline{y} - \underline{\mu} ^2 = \underline{y} - \underline{X}\underline{\beta} ^2 = \underline{\varepsilon}(\underline{\beta}) ^2$ $\hat{\underline{\beta}} = \arg \min_{\underline{\beta}} S(\underline{\beta}) = \arg \min_{\underline{\beta}} \underline{f} - \underline{R}\underline{\beta} ^2$
QR Method of LS	Let ¹ $\underline{Q}$ be orthonormal matrix ² $\underline{X} = \underline{Q}\underline{R}$ ³ $\underline{Q}^T \underline{y} = \begin{pmatrix} \underline{f} \in \mathbb{R}^p \\ \underline{r} \in \mathbb{R}^{n-p} \end{pmatrix} \Rightarrow S(\underline{\beta}) = \underline{Q}^T \underline{\varepsilon}(\underline{\beta}) ^2 = \dots = \underline{f} - \underline{R}\underline{\beta} ^2 + \underline{r} ^2$
QR Method (ctd.)	If $\underline{R}$ full rank: $\hat{\underline{\beta}} = \underline{R}^{-1} \underline{f} = \underline{R}^{-1} \underline{Q}^T \underline{y}$ $RSS = \underline{r} ^2 = \underline{\varepsilon}(\hat{\underline{\beta}}) ^2 = \underline{y} - \underline{X}\hat{\underline{\beta}} ^2$ ps: $\underline{f} = \underline{Q}^T \underline{y} \underline{Q}^T \underline{Q} = \underline{I}_p, \underline{Q}\underline{Q}^T \neq \underline{I}$

More Inference of LS:

Matrix Method of LS	$S(\underline{\beta}) = (\mathbf{y} - \mathbf{X}\underline{\beta})^T(\mathbf{y} - \mathbf{X}\underline{\beta}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\underline{\beta} + \underline{\beta}^T\mathbf{X}^T\mathbf{X}\underline{\beta}$	$\underline{\hat{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	$\mathbf{V}_{\underline{\hat{\beta}}} = Var[\underline{\hat{\beta}}] = (\mathbf{X}^T\mathbf{X})^{-1}\sigma^2$	
Influence / Hat Matrix A	$\mathbf{A} = \mathbf{Q}\mathbf{Q}^T = \mathbf{A}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$	# of (identifiable) parameters = $tr(\mathbf{A}) = tr(\mathbf{Q}\mathbf{Q}^T) = tr(I_p) = p$		
$Var[\mathbf{y}], Var[\mathbf{f}], \hat{\underline{\mu}}, \hat{\underline{\beta}}$	$Var[\mathbf{y}] = Cov(\underline{\varepsilon}, \underline{\varepsilon}) = \sigma^2 I_n$	$Var[\mathbf{f}] = \mathbf{Q}^T\mathbf{Q}\sigma^2 = I_p$	$\hat{\underline{\mu}} = \mathbf{X}\hat{\underline{\beta}} = \mathbf{A}\mathbf{y}$	$\hat{\underline{\beta}} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{y}$
$\mathbf{V}_{\hat{\underline{\beta}}}, \mathbf{V}_{\hat{\underline{\varepsilon}}}, \mathbf{V}_{\hat{\underline{\mu}}} \parallel \hat{\underline{\mu}} \text{ dist}$	$\mathbf{V}_{\hat{\underline{\beta}}} = Var[\hat{\underline{\beta}}] = \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2$	$\mathbf{V}_{\hat{\underline{\varepsilon}}} = Var[\hat{\underline{\varepsilon}}] = (\mathbf{I}_n - \mathbf{A})\sigma^2$	$\mathbf{V}_{\hat{\underline{\mu}}} = Var[\hat{\underline{\mu}}] = \mathbf{A}\sigma^2$	$\hat{\underline{\mu}}$ is Degenerate Normal (退化正太分布, Σ 不满秩)
$\mathbb{E}[\hat{\underline{\beta}}], \mathbb{E}[\hat{\underline{\varepsilon}}], \mathbb{E}[\hat{\underline{\mu}}] \parallel \hat{\underline{\varepsilon}} \text{ dist}$	$\mathbb{E}[\hat{\underline{\beta}}] = \underline{\beta}$ unbiased	$\mathbb{E}[\hat{\underline{\varepsilon}}] = \mathbf{0}$	$\mathbb{E}[\hat{\underline{\mu}}] = \underline{\mu}$ (unbiased)	$\hat{\underline{\varepsilon}}$ is Degenerate Normal (退化正太分布, Σ 不满秩)
If $\mathbf{X}^T\mathbf{X}$ is diagonal $\Rightarrow \mathbf{X}$ is called Orthogonal. $\Rightarrow$ In this Case: $\mathbf{V}_{\hat{\underline{\beta}}}$ diagonal $\Rightarrow \beta_i$ independent.				

Prediction | PI: Create a Prediction Matrix: $\mathbf{X}^*$ ($\mathbf{X}^*$ 形式和 $\mathbf{X}$ 保持一直, 数据变为按拟合后的每个点的数据) $\Rightarrow \hat{\underline{\mu}}^* = \mathbf{X}^*\hat{\underline{\beta}}, \hat{\underline{\mu}}^* \sim \mathcal{N}(\underline{\mu}^*, \mathbf{X}^*(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^*)^T\sigma^2)$

4.4 CI / Hypothesis Testing

Normality Assumption: Assume that $\underline{\varepsilon} \sim \mathcal{N}(0, \sigma^2 I_n) \Rightarrow \mathbf{y} \sim \mathcal{N}(\mathbf{X}\underline{\beta}, I_n\sigma^2) \quad \mathbf{z} \sim \mathcal{N}(\underline{\beta}, \mathbf{V}_{\hat{\underline{\beta}}}) \parallel$ 对于 Normal, LS $\hat{\underline{\beta}} = \text{MLE } \hat{\underline{\beta}}$

$\mathbf{V}_{Q^T\mathbf{y}}, E[Q^T\mathbf{y}] \parallel \hat{\sigma}^2, \hat{\mathbf{V}}_{\hat{\underline{\beta}}}$	$\mathbf{V}_{Q^T\mathbf{y}} = Var[Q^T\mathbf{y}] = I_n\sigma^2$	$E[Q^T\mathbf{y}] = E\left[\begin{pmatrix} 1 \\ \mathbf{r} \end{pmatrix}\right] = \begin{pmatrix} 1 \\ \mathbf{r} \end{pmatrix}\underline{\beta}$	$\hat{\sigma}^2 = \frac{ \mathbf{r} ^2}{n-p}$ (unbiased)	$\hat{\mathbf{V}}_{\hat{\underline{\beta}}} = \mathbf{R}^{-1}\mathbf{R}^{-T}\hat{\sigma}^2$
f, r Distribution	Known σ^2 : $\mathbf{f} \sim \mathcal{N}(\mathbf{R}\underline{\beta}, I_p\sigma^2) \quad \mathbf{r} \sim \mathcal{N}(\mathbf{0}, I_{n-p}\sigma^2)$		Unknown σ^2 :	
Test Statistic	$T_{known \sigma^2}(\mathbf{Y}) = \frac{1}{\sigma^2} \mathbf{r} ^2 = \frac{1}{\sigma^2}\sum_{i=1}^{n-p}r_i^2 \sim \chi_{n-p}^2$		$T_{unknown \sigma^2}(\mathbf{Y}) =$	

4.5 Assumptions || Diagnostics

Graphical Assessment

- Residuals vs Fitted (Check linear assumption): $(\hat{y}_i, e_i), e_i = y_i - \hat{y}_i, \hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$
 在 0 附近, 越靠 0 越好。 $\parallel$ 离 0 远: something missing in the model. $\parallel$ 有趋势: Not Constant Var. $\parallel$ ps: 点在整张图均匀分布 $\rightarrow$ constant variance
- Scale-Location (Check constant variance assumption): $(\hat{y}_i, \sqrt{|e_i^*|}), e_i^* = \frac{e_i}{s_e}$ 越靠近 0.82 越好. (around a constant value)
- Normal QQ-Plot (Check Normal Assumption): $(z_i, e_i^*), z_i = \Phi^{-1}(\frac{i-0.5}{n})$

5 Distribution Table

Distribution	PDF	Mean	Var	Other
Bernoulli $X \sim Bern(p)$	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n, p)$
Binomial $X \sim Bin(n, p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$X, Y \sim Bin(n, p), Bin(m, p)$ $X + Y \sim Bin(n + m, p)$
For $X \sim Bin(n, p), n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$, then $X \sim Poisson(\lambda)$		For $X \sim Bin(n, p), n \rightarrow \infty$ and pdf symmetric, $\Rightarrow X \sim N(np, np(1-p))$		
Geometric $X \sim Geom(p)$	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	$E[X^2] = 2/p^2$
Negative Binomial $X \sim NB(r, p)$	$\binom{r+x-1}{x}(1-p)^r p^x, r \geq 1$	$\frac{rp}{1-p}$	$\frac{rp}{(1-p)^2}$	或者 $p = 1 - q$, 注意!
Poisson $X \sim Po(\lambda)$	$e^{-\lambda} \cdot \frac{\lambda^x}{x!}$	λ	λ	$E[X^2] = \lambda^2 + \lambda$
For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X + Y \sim Po(\lambda_1 + \lambda_2)$				
Multinomial $X_1 + \dots + X_k = n$ $(X_1, \dots, X_k) \sim MN(n, p_1, \dots, p_k)$	$\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, 0 < p_i < 1$	np_i	$np_i(1-p_i)$	
Uniform $X \sim \mathcal{U}([a, b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$E[X^2] = \frac{b^2+ab+a^2}{3}$
Normal $X \sim \mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbf{Y} \sim N_k(\underline{\mu}, \Sigma), \underline{\mu}$ vector $f_Y(y_1, \dots, y_k) = \frac{e^{-\frac{1}{2}(\mathbf{y}-\underline{\mu})^T \Sigma^{-1}(\mathbf{y}-\underline{\mu})}}{\sqrt{(2\pi)^k \det(\Sigma)}}$
For $X, Y \sim N(\mu_1, \sigma_1^2), Po(\mu_2, \sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$E[X^n] = \frac{n!}{\lambda^n}$
Beta $X \sim Beta(\alpha, \beta), \alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)}$	$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $Beta(\alpha, \beta)$ has maximum probability, $x = (\alpha - 1)/(\alpha + \beta - 2)$				
Gamma $X \sim \Gamma(\alpha, \beta), \alpha, \beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha - 1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2}, \frac{1}{2})} (1 + \frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0, 1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution)				
When $n \rightarrow \infty$ it will converges to $N(0, 1)$. Cauchy Distribution: $X \sim t_1$ and its mean and var are undefined.				
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}}\Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0, 1)$, then $\sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$, then $X + Y \sim \chi_{n+m}^2$				
F $X \sim F(\lambda, v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2}, \frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1, n_2}, X^{-1} \sim F_{n_2, n_1}$

6 Review

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error
Def	$\mathbb{P}(Rej\ H_0 H_0\ is\ T)$	$\mathbb{P}(Fail\ Rej\ H_0 H_0\ is\ F)$

-	H_0 is T	H_0 is F
Fail to Rej H_0	Right	Type II (β) 与 size 有关
Rej H_0	Type I (α) 与 size 无关	Right (Power)

Significance Level: $\alpha = \mathbb{P}(Rej\ H_0|H_0\ is\ T)$

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2 \min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test: $\mathbb{P}(\text{Rej } H_0|H_0 \text{ is F}) = 1 - \beta = \mathbb{P}(Z \in C|\mu = \mu^*)$ · 1.Sample size ↑, Power ↑; 2. σ^2 ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4. α ↑, Power ↑.

Def of Confidence Interval (CI): $100(1 - \alpha)\%$ Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

(Convergence of r.v.) Name	Definition	written as	Example	
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$	CLT 最弱:	见下面
Convergence in Distribution	If $\forall \epsilon > 0, \lim \mathbb{P}(X_n - X > \epsilon) = 0$.	$X_n \xrightarrow{p} X$	WLLN 中等:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$. $\Rightarrow \bar{X}_n \xrightarrow{p} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \xrightarrow{a.s.} X$	SLLN 最强:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$, $Var[X_i] < \infty, \Rightarrow \bar{X}_n \xrightarrow{a.s.} \mu$

Central Limit Theorem (CLT): If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty$. $\Rightarrow \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$ 样本均值的分布会趋近于正态分布

7 Writting Format

Distribution: 产生新 distribution 的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}[(A \cap B)/B]$
p-value: p-value measures the plausibility that the data were generated under H_0 . **Low p-value:** low p-value suggests H_0 is incompatible with the data and hence likely to be false.
CI: The set of values (e.g. θ) are not rejected at significance level α under H_0 . If data are generated under H_0 , there is a probability $(1 - \alpha)$ for the true θ_0 to be within the $(1 - \alpha)100\%$ CI.
Hypothesis Testing: **Steps:** ¹ State H_0, H_1 ; ² Select Test Statistic T ; ³ distribution of T under H_0 ; ⁴ observed test statistic; ⁵ critical value / region, p-value, or CI; ⁶ conclusion. **Reject H_0 :** We would reject H_0 in favour of the H_1 . (So that ...) **Fail to reject H_0 :** We fail to reject H_0 . Hence, there is strong evidence against the null hypothesis.
MLE: Using (Compare) Newton's Method = (≠) Scoring Method: Because the second derivative of likelihood function doesn't (does) include the r.v.
MLE Inference Parameter $|\mathbb{P}(\theta < ?) = \alpha$ is false: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).
Geometric Comparison of $T_{Score}, T_{Wald}, T_{LR}$: ¹ T_{Score} (Slope Comparison): Represents the slope of the likelihood function at θ_0 ; A steeper slope indicates that the likelihood function is changing more rapidly at θ_0 , which implies a greater difference between $\hat{\theta}$ and θ_0 . ² T_{Wald} (Width Comparison): Compares the difference between $\hat{\theta}$ and θ_0 . ³ T_{LR} (Height Comparison): Compares the maximum likelihood value $L(\hat{\theta})$ and the likelihood value at $\theta_0, L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\hat{\theta}$ and θ_0 $T_{Score}, T_{Wald}, T_{LR}$ **Other** 部分公式可以近似的理由: This is due to $\hat{\theta}$ being a consistent estiamtor of θ_0 , and $J(\theta_0)$ being a consistent estimator of $I(\theta_0)$
Normal Linear Regression Assumptions: Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.
Linear Model: LS Estimation: $S(\beta)$ small: μ_i relatively close to the y_i ; Large: μ_i far from their corresponding y_i . || **Meaning of ϵ :** The random component of the model is intended to capture the fact that if we gathered a replicate set of data, the β_i (coefficient) would not change, but the Y_i would, as a result of different measurement errors. || **Why Use Least Square to estimate β :** Most of time, Linear Model can be a system of n equations with p parameters with $n > p$. In this case, the system is overdetermined and there is no exact solution. The least square method is used to find the best solution. **Geometric Interpretation of Linear Model****||LS Method:** The least Square Estimation of Linear Model amounts to finding the *orthogonal projection* of the n dimensional response data $\mathbf{y}$ onto the p dimensional linear subspace spanned by the columns of $\mathbf{X}$. **Geometric Interpretation of Linear Model****|| $\mathbb{E}[\mathbf{y}]$:** $\mathbb{E}[\mathbf{y}]$ lies in the space spanned by the columns of $\mathbf{X}$, | and the least square estimate seeks to find the point in this space that is closest to $\mathbf{y}$.